Project 2: The Librarian

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Repository: github.com/Rahma-sbei/CS496-project2-The-Librarian
Live app: the-librarian-6wzd.onrender.com (may take a minute to wake up)

Overview. The Librarian is a retrieval-augmented question answering service over the Streamlit documentation, its closed GitHub issues and its release notes (404 pages, 418 issues, 60 releases → 3,562 indexed chunks); it answers only from retrieved passages, cites the source behind every claim with `[n]` markers, and replies "I don't know" when the passages do not contain the answer. On our 112-question evaluation it answers correctly **89.7%** of the time with **98.3%** of claims supported by a retrieved passage, at **$0.00057** and **1.2 s** per question.

1. Architecture and model choices

Offline (run once): `build_corpus.py` clones the docs and pulls issues and releases through the GitHub API, stripping personal data; `index_corpus.py` cleans and chunks the text (≈250 words, code blocks never split), gives each chunk a dense meaning vector and a sparse BM25 vector, and upserts it into Qdrant. *Online (per question):* the question (optionally rewritten by the model) goes to keyword search and vector search (50 candidates each); the two lists are fused by reciprocal rank fusion; an optional filter keeps only docs, issues or releases; the top **10** chunks go to the generator, which answers with `[n]` citations or "I don't know". A FastAPI service and web page wrap this and log cost and latency; rate limits cap each visitor at 10 questions/minute and the service at 300/day.

Component	Choice	Justification
Embedding model	`BAAI/bge-small-en-v1.5` (384-dim, ONNX/fastembed)	Strong retrieval quality for its size on the MTEB benchmark while only 384 dimensions, so storage and search stay cheap. It runs on CPU with no GPU and no embedding API, which keeps the service free and under the 512 MB host limit.
Vector database	Qdrant Cloud (one collection, named `dense` + `bm25` vectors)	Stores both vectors per point and computes the BM25 word-rarity (IDF) weights server-side, so hybrid search is a single query. Point ids are derived from the document id and chunk number, so adding or replacing a document is one upsert with no rebuild — unlike our first flat-file index, which had to be rebuilt for every new document. Free tier keeps the app small.
Generative model	`gpt-4o-mini`, temperature 0	Follows the strict contract (cite every factual sentence; use nothing outside the passages; reply exactly "I don't know") reliably at $0.15/$0.60 per million input/output tokens. Retrieval, not the generator, is the accuracy ceiling (faithfulness is already 98%+), so a larger model buys little; `gpt-4.1-mini/nano` were available but did not justify the change.
Grader (judge) model	`gpt-4.1-mini`, temperature 0, JSON mode	A different and stronger family than the generator, which avoids self-preference bias when an LLM scores LLM answers. Re-grading the baseline moved accuracy only 86.2%→85.3%, confirming the headline is not inflated by a model grading its own family. It runs only during evaluation (≈$0.17 per 112-question run), never in the live service.

2. Optimization

Almost all per-question cost (85%) is input tokens, and almost all of the input is the passages; search itself is free (embeddings run locally, Qdrant is on a free tier). We measured both directions on the same 112 questions under the independent grader. **Accuracy:** sending 10 passages instead of 5 puts the gold passage in front of the model 87% of the time instead of 77% and raises accuracy **+4.4 points** at no latency cost and about $0.00025 more per question — this is the **deployed** configuration. A cross-encoder reranker (top 30 → best 5) lifts recall but *lowers* end-to-end accuracy 1.8 points and triples latency on a CPU, so it is not used.

Retrieval sent to the model	Accuracy (112)	Gold passage sent	Cost/question	p50 / p95 latency
5 passages (previous)	85.3%	77%	$0.000317	1.27 s / 2.84 s
10 passages (deployed)	**89.7%** (+4.4)	**87%**	$0.000569	1.19 s / 2.45 s
Rerank top 30 → best 5	83.5% (−1.8)	80%	$0.000289	3.07 s / 10.35 s

Cost (Week 8 target). The brief asked to cut cost or p95 latency by 50% within a 2-point accuracy loss. Sending only 2 passages cuts cost 53% ($0.00015) but, measured on the same set, drops accuracy about 9 points — well past the budget — so we did not adopt it. Two savings cost nothing: we skip the reranker (≈3.5 s/question avoided), and the service answers repeated questions from an in-memory cache with no search and no model call. Full before/after numbers are in `optimisation.md`.

3. Evaluation

Retrieval and generation are measured separately. The set is 100 answerable questions (each with a reference answer and one or more exact source passages) and 12 unanswerable questions about features Streamlit does not have. A question counts as "found at k" if any of the top k chunks contains one of its passages; the scripts recompute this for whichever collection they test.

Retrieval ablation. Fusing keyword and vector search beats either alone; the reranker helps the top ranks but

is too slow for an interactive service (times include the round trip to Qdrant Cloud).

Search	R@1	R@3	R@5	R@10	R@20	Doc R@5	MRR@10	p50 ms	p95 ms
Keyword only (BM25)	0.32	0.56	0.72	0.83	0.89	0.82	0.48	47	52
Vector only	0.47	0.67	0.70	0.82	0.85	0.79	0.58	55	61
Hybrid (RRF, used)	0.44	0.66	0.77	**0.87**	0.93	0.83	0.58	60	68
Hybrid + reranker	0.44	0.75	0.80	0.88	0.93	0.85	0.60	3,526	3,625

Generation (deployed: 10 passages, `gpt-4o-mini`, judged by `gpt-4.1-mini`). The judge splits each answer into claims, checks each against the passages and its citation, and grades it against the reference; an unanswerable question scores 1 only if the answer is "I don't know".

Generation metric	Result	Correctness by type	
Accuracy, all 112 questions (headline)	**89.7%**	conceptual (n=17)	94%
Correctness on the 100 answerable (yes=1, partial=0.5)	88.5%	error-message (n=13)	100%
Faithfulness: claims stated by a retrieved passage	98.3%	how-to (n=47)	83%
Answers in which every claim is supported	96.9%	multi-part (n=8)	100%
Citation accuracy: the cited passage states the claim	98.8%	version (n=15)	83%
Claims that carry a citation	99.6%	unanswerable (n=12)	100%
Answers citing a source number that does not exist	0.0%		
"I don't know" on the 12 unanswerable (want 100%)	100%	Gold passage among the	
"I don't know" on answerable questions (want 0%)	4.0%	10 chunks sent: 87%	

Cost model. Embeddings run on our CPU and vectors live on Qdrant's free tier, so search is free; the only per-question cost is the language model.

Configuration	Per question	Per 1,000	Per month @100/day	Per month @10,000/day
10 passages (deployed)	$0.00057	$0.57	$1.71	$170.82
2 passages (cost floor)	$0.00015	$0.15	$0.45	$44.86

Reading the numbers. When the right passage reaches the model it rarely strays beyond it (98.3% faithfulness, 98.8% citation accuracy); the remaining errors are mostly retrieval misses — the gold passage was among the 10 sent for 87% of questions, and most not-fully-correct answers never received it. "I don't know" works in both directions: 100% on unanswerable questions, 4% on answerable ones. Better retrieval would raise accuracy more than a better prompt. A full postmortem of 23 problems and fixes is in `postmortem.md`.